

The liver in haematologic oncologic disorders

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*Special considerations in the haem-onc patient are bolded.
*R (ratio) value = (ALT/ALT ULN) / (ALP/ALP ULN).

Elevated liver enzymes in the haem-onc patient

Hepatocellular injury (R ≥5)

Drug-induced liver injury
Sepsis
Viral infection
Ischaemic liver injury
Budd Chiari syndrome or splanchic thrombosis
Cytokine release syndrome

Sinusoidal obstruction syndrome
Autoimmune/other immune-mediated hepatitis
Cholelithiasis
Alcohol-associated hepatitis
Atypical thrombotic thrombocytopenic purpura

Mixed injury (R 2-5)

Drug-induced liver injury
Sepsis
Iron overload
Graft versus host disease
Haemophagocytic lymphohistiocytosis

Nodular regenerative hyperplasia
Sweet syndrome
Hepatitis-associated aplastic anemia
Cholelithiasis
Primary/metastatic hepatic neoplasm

Cholestatic injury (R ≤2)

Drug-induced liver injury
Sepsis
Iron overload
Graft versus host disease

Parenteral nutrition-associated liver disease
Coincidental PBC/PSC
Cholelithiasis
Granulomatous disease/amyloidosis

Sinusoidal obstruction syndrome

Sinusoidal obstruction syndrome, also known as **vaso-occlusive disease**, is caused by toxic injury to sinusoidal endothelial cells (by conditioning chemotherapy and/or radiation for haematopoietic stem cell transplantation [HSCT], or other hepatotoxins). This leads to a cascade of inflammation, procoagulation and fibrinolytic pathway activation, sinusoidal fibrosis and obstruction, hepatocyte necrosis, and ultimate portal hypertension, and may progress to acute liver failure.

- Damage to sinusoidal endothelial cells and activation of procoagulation, fibrinolytic and inflammatory cascades
- Sinusoidal dilation, congestion with erythrocytes, thrombus formation, and obstruction
- Sinusoidal and hepatic venule fibrosis, perivascular hepatocyte necrosis

Presentation

- Signs & symptoms:** Abdominal pain, ascites, peripheral edema, hepatomegaly, typically in first 21 days post-HSCT.
- Liver tests:** ↑ AST/ALT, ↑ Tbili, ↓ platelets.

Diagnosis

Clinical (modified Seattle, Baltimore, or European Bone Marrow Transplantation Society criteria may be used). Transjugular liver biopsy with HVPG measurement for high suspicion not meeting clinical criteria.

Liver biopsy

Sinusoidal hemorrhage/congestion, perivascular necrosis and fibrosis, and obliteration of small hepatic venules.

Treatment

Supportive care. Consider defibrotide (reduced progression to ALF and mortality in moderate-severe SOS) or TIPSS. May be an indication for transplantation.

Graft versus host disease

Graft versus host disease (GVHD) is a life-threatening multisystem disorder where the allograft's immune cells attack the host's body cells. Liver GVHD most commonly occurs in patients with skin and GI GVHD.

Acute GVHD occurs <100 days post-HSCT and is driven by the donor's T cells attacking host cells. **Chronic GVHD** occurs ≥100 days post-HSCT and is driven by both T cell and humoral immune mechanisms.

Presentation

- Signs & symptoms:** In the majority, accompanying skin (pruritic, painful maculopapular rash in acute GVHD, lichen-planus-like lesions in chronic GVHD) and GI manifestations (abdominal pain, nausea, vomiting, diarrhea, mucosal ulceration, anorexia).
- Liver tests:**
 - Acute GVHD:** ↑ TBili, and sometimes ↑ AST/ALT.
 - Chronic GVHD:** ↑ ALP/SGT, ↑ AST/ALT. TBili most often normal.

Diagnosis

Clinical if supported by skin and GI findings. Liver biopsy needed to document GVHD.

Liver biopsy

Biliary epithelial degenerative changes, centrilobular canalicular cholestasis, ductopenia.

Treatment

Corticosteroids first-line. Second-line options include calcineurin inhibitors and JAK inhibitors (e.g. ruxolitinib).

Nodular regenerative hyperplasia

Nodular regenerative hyperplasia (NRH) is characterized by diffuse micronodular transformation of the liver parenchyma without fibrosis, leading to non-cirrhotic portal hypertension.

NRH is associated with myeloproliferative/lymphoproliferative disorders, sickle cell disease, spherocytosis, common variable immunodeficiency and chemotherapy.

Presentation

- Signs & symptoms:** Ranges from asymptomatic to life-threatening portal hypertension complications, such as variceal bleeding.
- Liver tests:** AST, ALT, ALP and TBili normal to mildly elevated.

Diagnosis

Liver biopsy +/- HVPG determination.

Liver biopsy

Small nodules of regenerative hepatocytes with no or minimal fibrosis; internodular hepatocyte atrophy (reticulin stain shown). May be associated with obliterative portal venopathy.

Treatment

Treat the underlying disorder; address portal hypertension complications. NRH may be an indication for transplantation.

Haemophagocytic lymphohistiocytosis

Haemophagocytic lymphohistiocytosis (HLH) is a rare, life-threatening syndrome of cytolytic dysfunction of cytotoxic T cells and NK cells resulting in persistent antigen presentation, macrophage proliferation, hypercytokinemia, sustained inflammation, and multiorgan damage.

HLH may be **primary** (associated with a familial disorder or known genetic defect) or **secondary** (associated with triggers including infections, autoimmune conditions, acquired immunodeficiencies, immunosuppression, or malignancy).

Presentation

- Signs & symptoms:** Persistent fever, hepatosplenomegaly, rash, lymphadenopathy, neurologic dysfunction (out of proportion to expected HE), liver dysfunction, and coagulopathy. May progress to shock and multiorgan dysfunction. ALF as a presentation is rare, but possible.
- Liver tests:** ↑ AST/ALT, GGT levels, ↑ TBili.

Diagnosis

Most commonly used criteria:

- H-score** (Includes immunosuppression, fever, organomegaly, hypertriglyceridemia, hyperferritinemia, hypofibrinogenemia, AST, cytopenias, and haemophagocytosis on bone marrow aspirate).
- HLH 2004 criteria** (Includes fever, splenomegaly, hypertriglyceridemia, hyperferritinemia, hypofibrinogenemia, cytopenias, haemophagocytosis on bone marrow aspirate, high sIL-2R, and low/absent NK cell activity).

Liver biopsy

Sinusoidal haemophagocytosis, endotheliitis, parenchymal collapse, hepatocyte necrosis.

Treatment

Manage disease triggers; treatment options include high dose corticosteroids, etoposide (VP-16), ATG and/or cytokine-directed therapy (e.g. emapalumab, JAK inhibitors), chemotherapy.

Keywords: graft versus host disease; haemophagocytic lymphohistiocytosis; nodular regenerative hyperplasia; sinusoidal obstruction syndrome; venoocclusive disease.
Abbreviations: ALF, acute liver failure; ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate aminotransferase; ATG, anti-thymoglobulin; GGT, gamma glutamyl transferase; GI, gastrointestinal; GVHD, graft versus host disease; HE, hepatic encephalopathy; HLH, Haemophagocytic lymphohistiocytosis; HSCT, haematopoietic stem cell transplant; HVPG, hepatic vein-portal gradient; JAK, janus kinase; NK, natural killer; PBC, primary biliary cholangitis; PSC, primary sclerosing cholangitis; TBili, total bilirubin; TIPSS, transjugular intrahepatic porto-systemic shunt; ULN, upper limit of normal.
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Gastroenterologists and hepatologists are not infrequently called upon to assess patients with haematologic-oncologic disorders who present with concerns over their liver function. Hepatobiliary injury is frequently associated with significant morbidity and mortality in this population, with reported cases of acute liver failure and cirrhosis development.¹ For this reason and owing to the growing use of haematopoietic stem cell transplantation (HSCT), the diagnosis and management of liver complications is important. The differential diagnosis is broad and may initially be classified according to the pattern and severity of serum liver test abnormalities.² Frequent hepatobiliary conditions that can affect this population include drug-induced liver injury, neoplastic infiltration of the liver, sepsis-associated liver injury, sinusoidal obstruction syndrome (SOS), graft vs. host disease (GVHD), haemophagocytic lymphohistiocytosis (HLH), viral hepatitis, ischaemic hepatitis, Budd-Chiari syndrome and splanchnic thrombosis, and nodular regenerative hyperplasia (NRH), among others.

SOS typically manifests post-HSCT with hepatomegaly, abdominal pain, ascites, weight gain, peripheral oedema, transfusion-refractory thrombocytopenia, and jaundice. Defibrotide remains the only agent for the treatment of severe SOS and has been associated with improvement in survival.³ Among patients with severe, rapidly progressive SOS, transjugular intrahepatic portosystemic shunt insertion can be considered.⁴ GVHD, on the other hand, may have a different presentation in the acute (<100 days) vs. chronic (>100 days) setting. Acute GVHD occurs in up to 62% of patients post-allogeneic HSCT.⁵ While the liver is infrequently involved compared to other organs in acute GVHD, GVHD with liver involvement is associated with higher HSCT-related mortality.⁵ The hepatic manifestation often presents as cholestatic disease alongside cutaneous and gastrointestinal symptoms, with systemic corticosteroids (and other immunosuppression) as standard treatment. Ruxolitinib, a JAK2 inhibitor, may be employed in corticosteroid-refractory cases.⁵

NRH can develop in some patients with myelo- or lymphoproliferative disorders or those who have received high-dose chemotherapy. NRH can lead to non-cirrhotic portal hypertension and its complications, and may even be an indication for transplantation.⁶ HLH, conversely, is a multisystem condition characterized by an uncontrolled and ineffective hyper-inflammatory response with high mortality rates. It may be primary (associated with a familial disorder or known genetic defect) or secondary (associated with triggers including infections, autoimmune conditions, acquired immunodeficiencies, immunosuppression, or malignancy). Patients typically present with jaundice, persistent high-grade fever, pancytopenia, splenomegaly, evidence of haemophagocytes in the bone marrow and biochemical abnormalities (including hyperferritinaemia), but it may also present as acute liver failure.⁷ The current treatment of HLH is based on immunosuppression, cytotoxic agents and treating the underlying disorder.⁸

Primary and secondary hepatic involvement is also possible in haematologic malignancies, including lymphomas, myeloid sarcoma (chloroma), multiple myeloma, and Castleman disease.⁹ Suggestive findings of a haematologic neoplasm as the underlying cause of liver lesions include young age, abnormal bone marrow biopsy findings, unexplained fever, and night sweats. Abdominal imaging may show hepatosplenomegaly, focal liver lesions, liver parenchymal changes, extrahepatic masses in the porta-hepatis, vascular changes, as well as disseminated lymphadenopathy.

Finally, patients with haematologic-oncologic disorders and elevated serum aminotransferase activity warrant assessment for a variety of causes of viral hepatitis, as well as drug-induced liver injury. In allogeneic HSCT recipients, viraemic donors can transmit viral hepatitis B and C to recipients. In patients with chronic hepatitis B virus infection or previous exposure to the virus, the disease may reactivate after chemotherapy or HSCT. The risk of reactivation depends on many factors (patient, disease, viral, treatment choices).¹⁰

Abbreviations

GVHD, graft vs. host disease; HLH, haemophagocytic lymphohistiocytosis; HSCT, haematopoietic stem cell transplantation; NRH, nodular regenerative hyperplasia; SOS, sinusoidal obstruction syndrome.

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Conflict of interest

The authors of this study declare that they do not have any conflict of interest. Please refer to the accompanying ICMJE disclosure forms for further details.

Authors' contributions

Initial concept: GMH. Concept development: GGLC, MD, and GMH. Initial draft and revisions: MD, GGLD, DD, and GMH. Submission approval: MD, GGLD, DD, and GMH.

Supplementary data

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